

Energy, Work & Power



Work done by a constant force

- 1 A force of 40 N pushes a crate a distance of 6 meters along a horizontal floor, in the direction of motion. Find the work done.
 - 2 A rope pulls a sled with a force of 80 N at an angle of 30° above the horizontal, moving it 10 meters horizontally. Find the work done.
 - 3 A force of 25 N pulls a box a distance of 12 meters along a horizontal floor, in the direction of motion. Find the work done.
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Kinetic energy

- 4 Find the kinetic energy of a car of mass 1000 kg traveling at 25 m/s.
 - 5 Find the kinetic energy of a cyclist and bicycle with combined mass 90 kg traveling at 8 m/s.
 - 6 Find the kinetic energy of a ball of mass 0.2 kg traveling at 15 m/s.
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Gravitational potential energy

- 7 Find the gravitational potential energy gained by a 60 kg hiker climbing 150 meters, using $g = 9.8 \text{ m/s}^2$.
 - 8 Find the gravitational potential energy gained by a 5 kg object lifted 12 meters, using $g = 9.8 \text{ m/s}^2$.
 - 9 Find the gravitational potential energy gained by a 15 kg object lifted 8 meters, using $g = 9.8 \text{ m/s}^2$.
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The work-energy theorem

- 10 A ball of mass 0.5 kg is moving at 4 m/s when a resultant force acts on it, increasing its speed to 10 m/s. Find the work done on the ball, using the work-energy theorem.
- 11 A car of mass 900 kg is moving at 20 m/s when the brakes bring it to rest. Find the work done by the braking force, using the work-energy theorem.
- 12 A skateboard and rider of mass 60 kg is moving at 3 m/s when a force accelerates them to 9 m/s. Find the work done, using the work-energy theorem.

Conservation of mechanical energy

- 13** A ball of mass 2 kg is released from rest at the top of a smooth (frictionless) ramp of height 5 meters. Find its speed at the bottom, using conservation of energy and $g = 9.8 \text{ m/s}^2$.
- 14** A pendulum bob of mass 1 kg swings from a height of 0.8 meters above its lowest point. Find its speed at the lowest point, using $g = 9.8 \text{ m/s}^2$.
- 15** A skateboarder of mass 50 kg starts from rest at the top of a smooth (frictionless) ramp of height 3.2 meters. Find their speed at the bottom, using conservation of energy and $g = 9.8 \text{ m/s}^2$.

Energy loss due to friction or resistance

- 16** A skier of mass 70 kg slides down a slope, descending a height of 40 meters. Their speed at the bottom is measured at 20 m/s. Find the energy lost to friction and air resistance, using $g = 9.8 \text{ m/s}^2$.
- 17** A sled of mass 20 kg slides down a slope, descending a height of 10 meters. Its speed at the bottom is measured at 12 m/s. Find the energy lost to friction, using $g = 9.8 \text{ m/s}^2$.
- 18** A crate of mass 30 kg slides down a ramp, descending a height of 6 meters. Its speed at the bottom is measured at 8 m/s. Find the energy lost to friction, using $g = 9.8 \text{ m/s}^2$.

Power

- 19** A motor does 18000 J of work in 12 seconds. Find its power output.
- 20** A car engine produces a constant driving force of 2000 N while traveling at a constant speed of 18 m/s. Find the power output, using $P = Fv$.
- 21** A pump does 54000 J of work in 45 seconds. Find its power output.
- 22** A crane does 63000 J of work in 70 seconds. Find its power output.

A well-designed word problem: a cyclist climbing a hill

- 23** This question has two parts. A cyclist and bicycle have a combined mass of 80 kg. The cyclist rides up a hill, climbing a vertical height of 30 meters over a time of 60 seconds, starting and ending at the same speed of 6 m/s (so kinetic energy does not change). Air and rolling resistance do a total of 900 J of negative work against the cyclist during the climb. Use $g = 9.8 \text{ m/s}^2$. (a) Find the total work the cyclist must do against gravity and resistance combined. (b) Find the cyclist's average power output during the climb.